MENYAI: ADULT LITERACY ENHANCEMENT PLATFORM FOR RWANDA

A Kinyarwanda-Language Platform with Optional AI Tutoring

BSc. in Software Engineering

Lina IRATWE

Supervisor: Pelin Mutanguha

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CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Adult literacy is a pressing issue within Rwandan development, especially among those who did not get a chance to be educated when they were still young. Adult literacy in Rwanda is estimated at 78.8%, indicating that more than a fifth of people aged 15 and above are not able to read and write, with illiteracy being most prominent among rural and female sections of society.

Rwanda remains predominantly agricultural, with many families living in areas where poverty, scarce educational facilities, and the aftermath of past conflicts have blocked adults from learning opportunities. Consequently, adults who cannot read or write face real barriers as they miss out on better employment, cannot read medicine labels or health instructions, feel excluded from community discussions, and struggle to benefit from mobile banking or government digital services.

Adult literacy programs in Rwanda have traditionally meant showing up to a classroom, whether at a community center, church, or evening class run by local NGOs or the government. But the reality is harsh: classes happen at fixed times that don't match when people are free. After a long day farming or working, the walk to class feels impossible. Mothers can't find childcare. There aren't enough trained teachers. So people attend when they can, miss weeks at a time, and eventually stop coming.

AI-powered learning platforms delivered entirely in Kinyarwanda offer a fundamentally different pathway. A Kinyarwanda-language mobile platform with optional AI assistance addresses barriers by enabling adults to engage with educational content on their own schedules, in their native language, through an interface they can understand without requiring English or French literacy.

The platform's optional AI assistant is activated only when the learner taps an AI help button, which acts as a patient tutor that speaks Kinyarwanda, never judges, and repeats explanations as many times as needed. Learners control when they want AI assistance, reducing data costs and allowing independent study. English and French are introduced gradually as subjects to be learned through Kinyarwanda instruction, not as languages required to use the platform.

1.2 Problem Statement

Two existing initiatives come closest to addressing adult literacy challenges in Rwanda: UNESCO's ABC mobile literacy program and the Kolibri digital learning platform.

The ABC initiative leverages mobile phones for delivering literacy instruction but remains constrained by text-based SMS technology with no intelligent tutoring. ABC's implementation in Rwanda has been inconsistent, with content primarily in French rather

than Kinyarwanda, creating a fundamental barrier for learners whose first language is Kinyarwanda.

Kolibri's interface and content are not Kinyarwanda-first systems requiring learners to navigate in English or French, assuming literacy in languages most adult learners have not yet mastered. Kolibri lacks intelligent tutoring; learners cannot ask questions in Kinyarwanda or receive personalized explanations.

Both solutions fail to provide truly Kinyarwanda-centered learning experiences. Existing platforms operate primarily in English/French with minimal Kinyarwanda support, creating a paradox where adults who cannot read are expected to navigate platforms in languages they're still learning.

The proposed platform addresses these limitations by providing a completely Kinyarwanda-language interface with optional intelligent tutoring that speaks only Kinyarwanda. The optional AI assistant (activated by tapping an AI button when needed) can engage in Kinyarwanda dialogue, clarify misunderstandings, and provide contextual examples, but only when the learner chooses to activate this support.

1.3 Project's Main Objective

The overall aim is to develop a Kinyarwanda-language Adult Literacy Enhancement Platform with optional AI assistance, beginning with a focused pilot in the Kacyiru and Gikondo sectors of Kigali.

This platform will deliver culturally relevant content entirely in Kinyarwanda, combining multimedia instruction with an optional AI conversational assistant (activated via an AI button) that provides on-demand tutoring, answers questions, and offers personalized feedback all in Kinyarwanda. English and French are taught as subjects through Kinyarwanda instruction.

The optional AI assistant will be trained on practical applications connected to Rwandan adults' daily lives, always communicating in Kinyarwanda. Learners choose whether to activate AI support based on their needs and data availability, ensuring the platform remains fully functional without AI assistance.

List of Specific Objectives:

Objective 1: Build a Kinyarwanda-language mobile learning platform with optional AI assistance - Create a phone-based learning tool with a completely Kinyarwanda interface and an optional AI assistant (accessed via an AI button) that provides on-demand personalized tutoring in Kinyarwanda. English and French are introduced as lesson subjects, not platform languages.

Objective 2: Develop practical literacy lessons in Kinyarwanda - Design learning modules entirely in Kinyarwanda, teaching reading through everyday needs specific to urban

Rwanda, such as reading medicine labels, understanding work contracts, using mobile money, and participating in Umuganda meetings. The optional AI provides explanations in Kinyarwanda when learners activate it.

Objective 3: Test and measure real impact in Rwanda - Pilot with 20 participants across Kacyiru and Gikondo sectors, track completion rates and AI interaction patterns (among those who use AI), measure whether users can read better in Kinyarwanda. Document how optional AI tutoring affects outcomes compared to independent learning.

1.4 Research Questions

How effective is a Kinyarwanda-language mobile learning platform with optional AI assistance in improving functional literacy skills among adults with minimal literacy in urban Rwanda? This encompasses:

1. What usability features best support adults using a Kinyarwanda-only mobile learning platform?
2. How does optional AI conversational tutoring (activated on-demand) influence learner engagement and outcomes compared to using the platform without AI?
3. How does a Kinyarwanda-first approach (with English/French taught as subjects) affect learning effectiveness compared to multilingual interface systems?
4. How does linking literacy instruction to Rwandan contexts through Kinyarwanda content affect learning relevance and motivation?
5. What types of optional AI interactions in Kinyarwanda are most effective when learners choose to activate support?
6. To what extent does participation improve Kinyarwanda reading, writing, and numeracy skills in Kacyiru and Gikondo sectors?
7. What factors influence whether learners choose to activate optional AI assistance versus learning independently?

1.5 Project Scope

Geographical Scope

This project focuses exclusively on two sectors in Kigali: Kacyiru Sector and Gikondo Sector. These represent different faces of urban Rwanda while remaining manageable for close monitoring.

Kacyiru Sector: Government offices, international organizations, middle-class residential areas. Adults work as domestic workers, security guards, or informal service providers needing literacy to navigate formal employment, read contracts, and understand government notices.

Gikondo Sector: Kigali's industrial hub with factories, workshops, and informal settlements. Adults work in manufacturing, construction, or small-scale trading, needing literacy for safety instructions, product labels, business transactions, and health services.

Target User Sample

The project will recruit 20 adult participants: 10 from Kacyiru and 10 from Gikondo, who:

- Are between 18 and 55 years old
- Can read fewer than 30 simple words in Kinyarwanda or any language
- Speak Kinyarwanda fluently as their primary language
- Have minimal smartphone experience
- Volunteer for three months
- Own a basic smartphone or can borrow one

Technical Scale

The platform will launch with three core learning modules delivered entirely in Kinyarwanda to ensure accessibility for adults with minimal literacy skills. These modules are structured progressively to build foundational reading, numeracy, and practical life competencies relevant to urban Rwandan contexts.

The first module, **Basic Literacy in Kinyarwanda**, consists of 20 lessons focused on introducing learners to Kinyarwanda letters, phonetic sounds, and simple words. Instruction emphasizes audio-visual reinforcement to support learners who are still developing reading confidence. Within this module, the optional AI assistant can pronounce words and explain letter sounds in Kinyarwanda when activated by the learner. English and French vocabulary are gradually introduced as lesson content, but they are taught through Kinyarwanda explanations to ensure that learners are not required to understand foreign languages in order to progress.

The second module, **Everyday Numbers**, contains 15 lessons covering counting, basic arithmetic, money calculations, and price interpretation. All instruction is delivered in Kinyarwanda and contextualized using familiar scenarios such as local market transactions and Rwandan currency. When learners choose to activate the AI assistant, it provides step-by-step problem-solving explanations in Kinyarwanda, breaking down calculations in simple, practical terms aligned with daily economic activities.

The third module, **Practical Life Skills**, also includes 15 lessons and focuses on functional literacy applications. Learners practice reading medicine labels, interpreting work safety signs, and understanding civic information relevant to Kigali. The optional AI assistant, when

activated, explains unfamiliar terms in Kinyarwanda and relates them directly to real workplace and community contexts to enhance comprehension and retention.

Each lesson is designed to last approximately 10–12 minutes, ensuring that content remains manageable for adult learners with limited study time. Across the three modules, the platform delivers a total of 50 structured Kinyarwanda lessons, supplemented by on-demand AI tutoring for learners who choose to use conversational support. This structure balances independent multimedia learning with optional personalized assistance, allowing learners to control their level of AI interaction based on need and data availability.

Devices and Technology

Works on affordable Android phones (50,000 - 80,000 RWF), common in Kigali Tecno, Infinix, Itel, and basic Samsung. Requires ~300MB storage, uses mobile data only when learners activate the optional AI assistant. Learners who don't activate AI complete lessons using Kinyarwanda audio and visual content without data usage. Those using AI need periodic internet through mobile data or WiFi. Lesson content cached for offline review.

Language Approach

The platform uses Kinyarwanda exclusively for all interface elements, navigation, instructions, and AI communication. The optional AI speaks only Kinyarwanda when activated. English and French are taught as subjects through Kinyarwanda instruction. Learners never need English or French to navigate, access features, or communicate with AI.

Success Metrics

- 12 out of 20 participants are still active after three months
- Each learner completes 25+ lessons
- Among AI users: 2-3 AI activations per learning session
- Measurable Kinyarwanda literacy improvement: reading 15-20 more words
- Real-world application: Half report using new Kinyarwanda literacy skills practically
- Clear insights on optional AI effectiveness and usage patterns
- Data showing whether the Kinyarwanda-only interface improved accessibility

1.6 Significance and Justification

This platform presents a context-sensitive and practical solution for adult learners in Kigali's Kacyiru and Gikondo sectors, including domestic workers who need to read employment contracts, factory workers who must understand safety instructions, and parents who wish to support their children's education. These adults require a learning system that aligns with

their daily realities, accommodates their schedules, and communicates in a language they fully understand.

The Kinyarwanda-only design directly addresses a structural barrier that has limited the effectiveness of many previous literacy interventions. It is pedagogically inconsistent to expect adults who cannot yet read to navigate digital platforms in unfamiliar languages. By ensuring that all interface elements, instructions, and learning interactions are delivered in Kinyarwanda, the platform removes the cognitive burden of simultaneously learning digital navigation, foreign-language terminology, and basic literacy skills. Learners are therefore able to concentrate exclusively on strengthening their reading abilities in a language they already speak fluently.

The inclusion of optional AI tutoring further enhances flexibility and learner autonomy. Adults can access conversational support in Kinyarwanda whenever they need clarification or additional explanation. For example, a domestic worker in Kacyiru may engage with the AI assistant in the evening to better understand employment-related terminology, while a market vendor in Gikondo may request step-by-step explanations for financial calculations relevant to daily sales. At the same time, learners who prefer independent study or who wish to conserve mobile data can complete lessons using the platform's audio-visual content without activating AI support. This learner-controlled approach respects varying technological comfort levels and economic constraints.

Beyond individual literacy gains, the platform has broader developmental implications aligned with Rwanda's Vision 2050. Functional literacy in Kinyarwanda strengthens civic participation and social inclusion. Adults who can read public notices, interpret development plans, and understand voting instructions are better positioned to engage meaningfully in community activities such as Umuganda and local governance processes. Literacy thus shifts individuals from passive recipients of information to active participants in civic and economic life.

For women in particular, native-language literacy can be transformative. The ability to independently read official documents, health information, or financial records reduces reliance on others and enhances personal agency. By eliminating foreign-language barriers within the learning interface, the platform fosters confidence and reduces intimidation, creating a supportive environment in which adults can build foundational literacy skills with dignity and autonomy.

1.7 Research Budget

This section explains costs for this small-scale pilot in two Kigali sectors. Only items requiring payment with no free alternative are included.

Research Budget (20 Participants in Kacyiru and Gikondo)

Item	Description	Quantity	Total Cost (RWF)
Participant transport	Local transport reimbursement	20 participants	60,000
Refreshments	Light refreshments during sessions	20 participants	40,000
Printing/stationery	Consent forms, assessment sheets	Lump sum	25,000
Mobile data vouchers	For those using optional AI	20 participants	80,000
AI service costs	API usage for conversational tutoring (3 months)	Cloud service	120,000
Cloud hosting	Platform hosting and data storage	3 months	60,000
Device support	Loaner smartphones	5 devices	75,000
Facilitation	Community assistant support	2 facilitators	80,000
Contingency	Unforeseen expenses	—	20,000
Total Budget			**560,000 RWF**

The budget is suitable as it:

- Focuses on unavoidable, payable costs
- Matches 20-participant pilot scale in two Kigali sectors
- Includes realistic AI service costs for optional use
- Supports data collection and evaluation
- Aligns with the 3-month academic project timeframe in Rwanda

1.8 Research Timeline

This study will be conducted over three months in the Kacyiru and Gikondo sectors, implementing key phases in a structured sequence. The Gantt chart shows how components align with time, ensuring development, AI integration, content preparation, piloting, and evaluation are realistically scheduled.

Gantt Chart: Research Timeline

Project Activity	Month 1	Month 2	Month 3
Literature review and proposal refinement	●		
Requirements gathering and user needs analysis	●		
Platform design (UI/UX for low-literacy, Kinyarwanda-only)	●	❖	
Optional AI assistant integration and Kinyarwanda training	❖	●	
Platform development (Kinyarwanda-only MVP)	❖	●	
Content development (Kinyarwanda multimedia modules)	❖	●	
Internal testing and AI refinement		❖	
Participant recruitment and onboarding		❖	

Pilot implementation (20 participants in Kigali)		❖	●
Data collection (assessments, AI interaction logs)			●
Data analysis and evaluation			❖
Report writing and documentation			●

The timeline allows flexibility for iterative improvements to both the platform and Kinyarwanda AI tutoring capabilities while ensuring completion within the allocated timeframe in Rwanda.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This literature review focused on finding research about actual software platforms and mobile applications designed for adult literacy, with particular attention to AI-powered educational technologies and applications in contexts similar to Rwanda. The search concentrated on platforms using intelligent tutoring systems, conversational AI, and adaptive learning technologies, particularly any implementations in African or low-resource contexts with attention to native language instruction.

Searches concentrated on Google Scholar, ERIC (education research), and IEEE Xplore (software implementation). Terms included 'AI tutoring system' with 'adult literacy,' 'conversational AI for education,' 'intelligent tutoring systems,' 'native language instruction,' and 'mobile learning' combined with 'Rwanda,' 'Africa,' and 'resource-constrained environments.'

Research prioritized papers after 2018 because AI technology, particularly conversational AI and NLP, has evolved extraordinarily rapidly. Older research about rule-based systems doesn't reflect current large language model capabilities. The focus was on finding research describing software and AI implementations—technical decisions about AI integration, NLP for local languages, deployment challenges, user interactions with AI tutors, and features supporting literacy in African contexts.

2.2 Historical Background

Adult illiteracy in Rwanda traces back to colonial education systems that educated only a tiny elite. When Rwanda gained independence, literacy campaigns used volunteer teachers and evening classes, but failed because they required too many teachers, and expected exhausted workers to attend using children's textbooks in foreign languages. The 1994 genocide destroyed the education infrastructure and left a generation focused on survival. While Rwanda has since achieved impressive progress in children's education, adults from that period mostly never caught up.

Late 2000s mobile learning experiments focused on SMS-based education, which didn't help people who couldn't read messages. The shift came around 2015 when affordable smartphones enabled audio-visual instruction—phones could speak local languages, show videos, use pictures. Pilot projects in Kenya and Tanzania demonstrated that adults could learn through sound and images.

The emergence of conversational AI and large language models from 2020 onwards created new possibilities for personalized tutoring. Unlike static content or SMS, modern AI can engage in genuine dialogue in multiple languages, answer questions, adapt explanations, and provide responsive tutoring. However, most AI educational technologies have been developed for well-resourced contexts with literate populations, leaving a gap in understanding how conversational AI can support adult literacy in native languages in contexts like Rwanda.

Rwanda's vision of becoming a digital knowledge economy assumes citizens can read notices, use online services, navigate smartphone apps—but these don't match reality for the quarter of adults who remain functionally illiterate, especially in a trilingual environment where official business happens in English and French while most people only comfortably speak Kinyarwanda. This research emerges from that convergence: persistent literacy gaps in Rwanda, widespread smartphone ownership, improved AI tutoring technology, and knowledge from decades of failed programs about what doesn't work—particularly the failure to leverage learners' native language fluency as the foundation for literacy instruction.

2.3 Overview of Existing Systems

When looking for digital literacy platforms operating in Rwanda, the landscape is sparse. Most educational technology focuses on children in formal schools, not adults. The few systems targeting adult learners reveal consistent gaps, particularly the absence of AI tutoring and native language primacy.

Ubumenyi (Rwanda's National Learning Platform)

Ubumenyi is primarily web-based, requiring a consistent internet connection. Adult education content is mostly PDF documents of text-heavy materials designed for classroom use. There's minimal audio support, limited Kinyarwanda content, and crucially, no AI tutoring

capabilities. Learners cannot ask questions in Kinyarwanda, request clarifications, or receive personalized explanations. The platform works for literate students, but for adults in Kacyiru or Gikondo who can barely read and need responsive Kinyarwanda tutoring, Ubumenyi provides little support.

Eneza Education

Eneza delivers content through SMS and USSD for basic phones but focuses on school curricula. The SMS format offers no conversational AI support. There's no intelligent assistant to answer questions, provide alternative explanations in Kinyarwanda, or adapt instruction. An adult learner in Kigali wouldn't benefit from school-oriented content or non-adaptive delivery.

Kolibri

Kolibri is open-source for offline learning, but its content primarily consists of Khan Academy videos and school curriculum materials. The interface is not Kinyarwanda-first—requiring navigation in English or French. It lacks intelligent tutoring—learners cannot ask questions in Kinyarwanda, request clarifications, or receive personalized explanations. There's no AI to adapt to difficulty, provide alternative Kinyarwanda explanations, or offer encouragement.

UNESCO's Literacy ABC App

UNESCO's ABC app teaches basic literacy through audio-supported lessons for adults. However, implementation in Rwanda has been limited. More critically, ABC lacks AI tutoring capabilities; it delivers static lessons without responding to learner questions, adapting explanations, or providing personalized feedback in Kinyarwanda. When a learner struggles, there's no intelligent assistant to offer alternative explanations, break down material differently, or encourage. Content feels generic, with examples that don't adapt to Rwandan contexts or individual learner needs.

What's Missing

Consistent gaps emerge: First, a fundamental absence of AI-powered personalized tutoring in native languages; none of these systems can engage in Kinyarwanda dialogue with learners, answer specific questions, or adapt instruction. Second, language localization remains superficial, with minimal Kinyarwanda materials, and no AI capability to explain concepts in a learner's native language. Third, platforms lack sustainable local implementation and deep cultural contextualization to Rwandan realities.

An adult in Kacyiru or Gikondo who speaks Kinyarwanda fluently, can't read, owns a basic smartphone, and needs literacy for daily urban life has no existing solution providing responsive AI tutoring in Kinyarwanda tailored to their context. The specific combination of AI-powered conversational tutoring in Kinyarwanda, native-language-first interface design,

adult-relevant content connected to Rwanda, and sustainable local implementation hasn't been packaged together in one accessible platform for Rwanda's urban adult learners.

2.4 Review of Related Work

Several research studies examined digital literacy interventions and AI-powered educational technologies in contexts relevant to Rwanda.

Human-computer interaction and mobile learning studies demonstrate that interface design plays a critical role for adults with minimal literacy. Research shows icon-based navigation with spoken native language instructions is more effective than text-heavy menus. However, conversational AI has transformed this landscape—rather than relying solely on careful icon design, modern systems can provide natural language interaction where learners ask questions in their own words (Kim et al., 2010; Patel et al., 2010).

Research on AI tutoring systems demonstrates that conversational interfaces in native languages can significantly reduce barriers for low-literacy learners. Studies of intelligent tutoring systems in developing countries show that learners who can ask questions in their native language and receive spoken responses demonstrate higher engagement and better outcomes compared to static content delivery. AI's ability to rephrase explanations in the native language, provide contextual examples, and adapt difficulty creates more supportive environments than traditional digital content (UNESCO, 2021; World Bank, 2019).

Participatory design studies in African contexts highlight that AI systems must be carefully adapted to local languages and cultural contexts. Research on NLP for low-resource languages shows that successful implementation requires careful attention to code-switching patterns, cultural reference points, and specific language use in everyday contexts (Bidwell et al., 2014; Winschiers-Theophilus et al., 2012).

Rwanda-specific research examining technology usage among informal sector workers in Kigali found that although smartphone ownership was widespread, digital confidence remained low. Participants expressed fear of mistakes, embarrassment about seeking help, and anxiety about technology (Ndahiro, 2021). This finding is particularly relevant for AI-powered tutoring—an AI assistant providing patient, non-judgmental responses in learners' native language, never expressing frustration at repeated questions, offering encouragement, can address psychosocial barriers more effectively than human tutors, who learners may fear disappointing.

2.5 Summary of Reviewed Literature

The reviewed literature demonstrates that effective digital literacy systems for low-literacy adult learners must integrate technical usability and AI-powered personalized support in native languages. Studies show that while interface design choices such as audio-first instruction and familiar navigation are important, conversational AI has created new

possibilities for supporting learners through natural language interaction in their native language.

Research on intelligent tutoring systems confirms that AI-powered technologies can provide adaptive, personalized instruction in learners' native languages that responds to individual needs in ways static content cannot. The ability to ask questions in one's native language, receive clarifications, and get explanations rephrased differently significantly enhances learning outcomes, particularly for adults with minimal formal education.

The literature highlights that social, emotional, and psychological factors are critical. Rwanda-focused studies reveal that fear of mistakes, embarrassment about low digital skills, lack of supportive environments discourage adult learners—challenges that AI tutoring systems in native languages are uniquely positioned to address through patient, non-judgmental, always-available conversational support.

Overall, the literature provides a strong foundation for the proposed Kinyarwanda AI-powered system design, justifying an approach integrating inclusive interface design with conversational AI tutoring in learners' native language to enhance adult digital literacy outcomes in Rwanda.

2.6 Strengths and Weaknesses of Existing Systems

Existing platforms demonstrate important strengths in expanding educational access through mobile delivery and partnerships. These systems leverage widespread smartphone ownership and standardized content distribution at scale.

However, reviewed systems fall short in addressing specific cognitive, emotional, and usability needs of adults with minimal literacy in Rwanda. Most platforms were designed for school-aged learners or formally educated users, resulting in interfaces assuming baseline reading confidence and independent navigation skills.

Most critically, none of the existing systems deployed in Rwanda incorporates AI-powered conversational tutoring in Kinyarwanda. Platforms provide content but no intelligent assistant to answer questions in learners' native language, adapt explanations, or provide personalized feedback. This absence of AI tutoring in native languages represents the most significant gap in current adult literacy solutions for Rwanda.

Systems also lack deep Kinyarwanda language integration, culturally relevant content specific to Rwandan urban contexts, and psychosocial support mechanisms that AI conversational interfaces in native languages can uniquely provide. These gaps highlight the need for a system integrating low-literacy HCI principles, AI-powered conversational tutoring in Kinyarwanda, localized multimedia content, and structured human facilitation.

2.7 General Comment and Conclusion

Overall, the reviewed literature and existing systems demonstrate that digital platforms have strong potential to support literacy development in Rwanda, particularly when enhanced with AI-powered personalized tutoring in native languages. However, evidence shows that current solutions prioritize content delivery and scalability over the lived realities of adults with minimal literacy, especially their limited digital confidence, anxiety around technology, and need for responsive, conversational learning support in their native language.

While initiatives such as Ubumenyi, Kolibri, Eneza Education, and UNESCO-supported tools provide valuable foundations, they do not integrate AI-powered conversational tutoring in Kinyarwanda, adult-centered design, native-language-first interfaces, supportive onboarding, or confidence-building mechanisms. The emergence of accessible AI technologies creates new possibilities for addressing these gaps.

This gap in both research and practice justifies the need for a tailored, AI-driven solution that combines intuitive interface design, conversational AI tutoring in Kinyarwanda, localized multimedia content relevant to urban Rwandan contexts, and human facilitation to effectively support adult learners in Kacyiru and Gikondo sectors of Kigali, with potential for expansion across Rwanda.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the system analysis and design of the proposed Kinyarwanda AI-powered adult literacy enhancement platform for Rwanda, focusing on how software and AI design choices can directly address challenges faced by adults with minimal literacy in Kacyiru and Gikondo sectors of Kigali. The design approach combines user-centered design, participatory research, and Agile methodology, ensuring the system evolves through continuous feedback from real users in Rwanda.

Rather than prioritizing technical complexity, the design emphasizes simplicity, clarity, emotional safety, and intelligent responsiveness—all in Kinyarwanda. The integration of optional conversational AI tutoring in learners' native language represents the core innovation—providing personalized, always-available support that adapts to individual learning pace, answers questions in Kinyarwanda, and offers encouragement, but only when learners choose to activate this support via an AI button. The design recognizes that confidence, usability, native language accessibility, and optional responsive AI interaction are as important as instructional content for adult learners in Rwanda.

3.2 Research Design

The system is developed using an Agile, iterative model, allowing features—particularly Kinyarwanda AI tutoring capabilities—to be introduced gradually, tested with users in Kigali, and refined based on observed difficulties. This approach is especially suitable for Rwanda's

adult literacy contexts, where user needs and optimal AI interaction patterns in Kinyarwanda often emerge during actual use.

Each iteration focuses on a small set of features—such as onboarding, lesson navigation in Kinyarwanda, optional AI question-answering in Kinyarwanda, or adaptive assessments ensuring steady progress while minimizing user overwhelm. The AI assistant itself undergoes continuous refinement based on actual learner questions in Kinyarwanda and interaction patterns observed during the Kacyiru and Gikondo pilot.

Proposed Development Model:

User Needs → Design → Prototype → User Testing → AI Refinement → Feedback → Improvement

This cycle ensures that adult learners in Rwanda with limited literacy actively shape both the platform design and the Kinyarwanda AI tutoring behaviors rather than adapting to rigid software assumptions. The AI's Kinyarwanda conversational responses, language choices, and explanation strategies are continuously adjusted based on what actually helps Rwandan learners understand concepts.

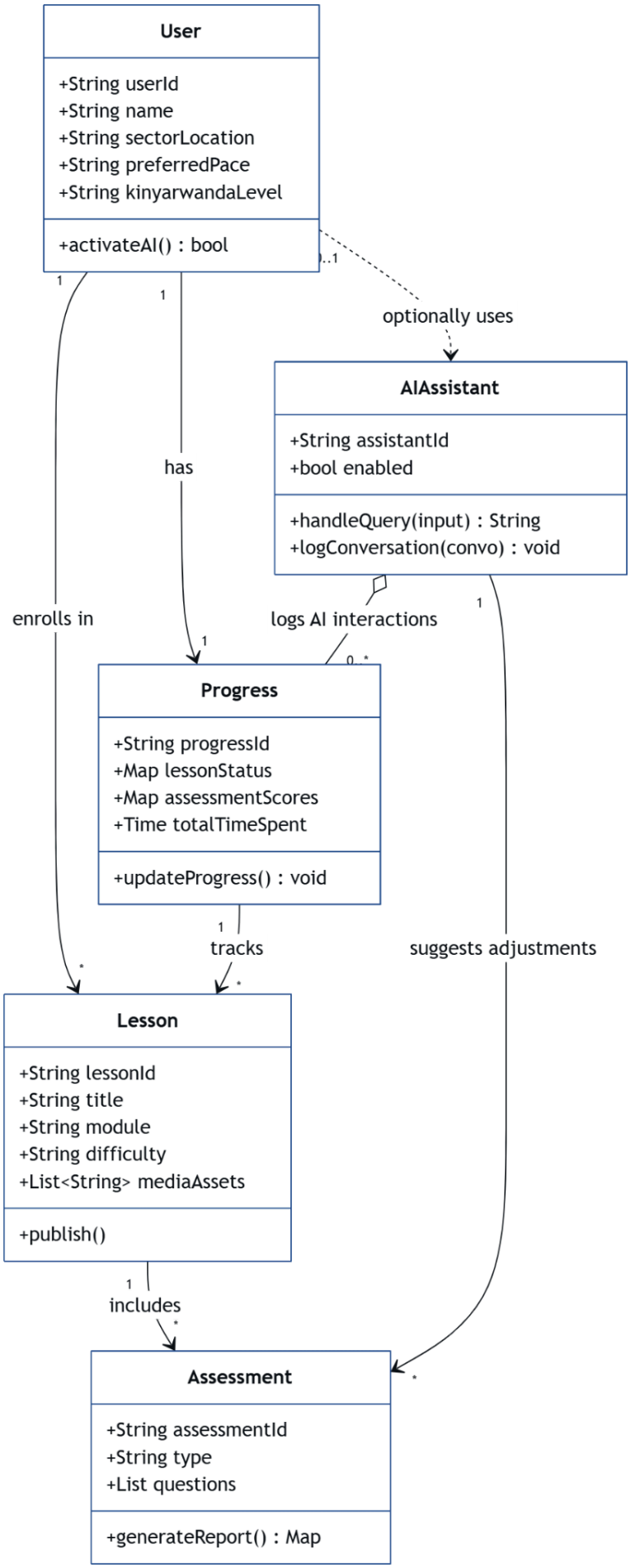
3.3 Class Diagram

The class diagram outlines the primary system components and illustrates how they interact to enable structured learning and optional AI-supported personalization in Kinyarwanda. The design intentionally maintains a minimal number of classes to ensure clarity, scalability, and maintainability, while clearly separating responsibilities across the system. Each class represents a distinct functional unit that contributes to delivering adaptive and user-centered literacy instruction.

At the center of the system is the **User** class, which stores essential learner profile information. This includes the learner's name, sector location within Kigali (e.g., Kacyiru or Gikondo), preferred learning pace, and current Kinyarwanda literacy level. The User class serves as the reference point for tracking progress, tailoring content, and managing AI interactions.

The **Lesson** class contains the structured educational content delivered in Kinyarwanda. It manages multimedia materials such as audio instructions, visual aids, and interactive exercises, along with metadata such as lesson difficulty level and module classification. This class ensures that instructional content remains organized, modular, and easily expandable as new lessons are added.

The **Progress** class is responsible for monitoring learner advancement throughout the platform. It records lesson completion status, assessment scores, time spent per lesson, and overall learning trajectory. This data supports both performance tracking and adaptive learning adjustments.


User
+String userId
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+String sectorLocation
+String preferredPace
+String kinyarwandaLevel
+activateAI() : bool
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+bool enabled
+handleQuery(input) : String
+logConversation(convo) : void
1
has
1
enrolls in
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logs AI interactions
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+Map assessmentScores
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tracks
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suggests adjustments
Lesson
+String lessonId
+String title
+String module
+String difficulty
+List<String> mediaAssets
+publish()
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includes
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Assessment
+String assessmentId
+String type
+List questions
+generateReport() : Map
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The **AIAssistant** class manages the optional conversational tutoring component. It handles learner queries, generates explanations in Kinyarwanda, provides adaptive feedback, and responds contextually to user input. Importantly, this component is activated only when the learner chooses to engage the AI feature, ensuring that personalization remains user-controlled and data-efficient.

Finally, the **Assessment** class evaluates learners' literacy and numeracy skills through structured exercises and quizzes. It generates performance data that can inform content progression and, when applicable, guide AI-driven adjustments to explanation depth or difficulty level. Together, these classes form a cohesive architecture that supports structured instruction, learner progress tracking, and optional intelligent tutoring within a streamlined and scalable system design.

AI Interaction: Logs all AI conversations in Kinyarwanda for analysis and system improvement (only for learners who activate AI)

This structure supports optional AI-powered personalized learning while remaining lightweight and scalable for deployment across Rwanda.

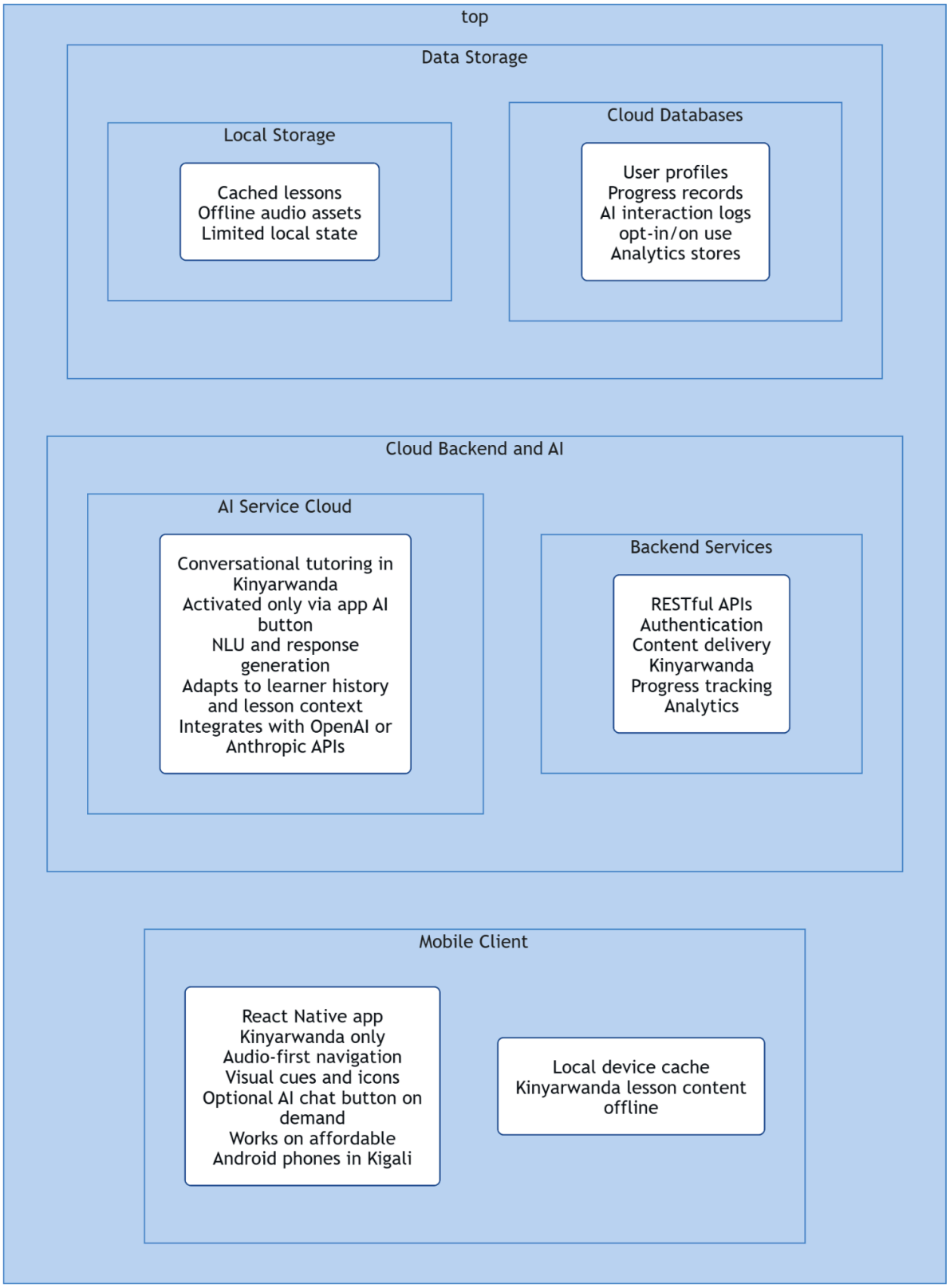
3.4 System Architecture

The system architecture follows a cloud-connected mobile-first model specifically designed for Rwanda's urban mobile network infrastructure. The architecture is divided into three clear layers: mobile client, cloud-based AI and backend services, and data storage.

The frontend consists of a Kinyarwanda-only mobile application with audio-first navigation, visual cues, and an integrated optional AI chat interface (activated via button), designed for learners with minimal literacy. The application works on affordable Android phones common in Kigali and uses mobile data only when learners choose to activate the optional AI assistant, leveraging Rwanda's relatively good urban network coverage in Kacyiru and Gikondo sectors.

The backend exposes RESTful APIs that manage authentication, Kinyarwanda content delivery, progress tracking, and analytics. Critically, it integrates with cloud-based AI services (such as OpenAI API or Anthropic Claude API) to power the optional conversational tutoring system in Kinyarwanda. The AI service handles natural language understanding in Kinyarwanda, generates contextually appropriate responses in Kinyarwanda, and adapts explanations based on learner interaction history, but only when activated.

Data storage combines cloud-based databases for user profiles, progress tracking, and AI interaction logs (for those who use AI), with local device caching for Kinyarwanda lesson content.



The mobile frontend is implemented as a React Native application designed specifically for low-literacy adult learners in Rwanda. The interface operates entirely in Kinyarwanda and prioritizes voice-based guidance and visual navigation elements rather than text-heavy menus. Audio prompts, clear icons, and structured lesson flows ensure that learners with

minimal reading skills can navigate independently. An optional AI chat button is embedded within the interface, allowing learners to activate conversational tutoring support only when needed. This design ensures accessibility while maintaining user control over AI engagement and data usage.

The AI service layer operates in the cloud and provides conversational tutoring in Kinyarwanda. This component is activated strictly on demand when a learner taps the AI button within the mobile application. Once activated, the system processes user queries, generates contextually appropriate explanations in Kinyarwanda, and adapts responses based on lesson context and learner progress. By keeping the AI service cloud-based, the platform avoids heavy on-device processing while enabling scalable, intelligent tutoring capabilities that remain optional and user-controlled.

The backend API is developed using Node.js and serves as the core logic layer of the system. It manages user authentication, lesson delivery, Kinyarwanda content management, learner progress tracking, and communication between the mobile frontend and the AI service layer. The backend ensures secure data handling and structured interaction between components while maintaining modularity for future expansion beyond the pilot phase in Kacyiru and Gikondo sectors.

The system uses a cloud-based database, specifically Firebase, to store user profiles, lesson progress data, and AI interaction logs for learners who choose to activate the AI assistant. This centralized storage approach enables real-time synchronization, secure authentication, and structured analytics. AI interaction logs are recorded only for users who engage the conversational assistant, allowing analysis of tutoring patterns while respecting learner choice and data efficiency.

To address Rwanda's mobile data constraints, the system incorporates local caching using SQLite on the learner's device. Static Kinyarwanda lesson content, including audio and visual materials, is stored locally after initial download. This allows learners to review lessons offline without requiring continuous internet connectivity. Only conversational AI interactions require live cloud connectivity, ensuring that independent learners can complete the majority of their learning without incurring ongoing data costs.

Overall, this architecture carefully balances optional AI-powered personalization with practical mobile infrastructure realities in Rwanda. By caching static Kinyarwanda content locally while keeping conversational AI features cloud-based and activated only when needed, the system supports intelligent tutoring without imposing unnecessary connectivity requirements. This hybrid model ensures scalability, affordability, and accessibility within Kigali's urban mobile environment while remaining adaptable for broader deployment across Rwanda.

3.5 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) defines how data is organized, stored, and interconnected within the system. It ensures data integrity, consistency, and scalability while supporting structured lesson delivery and optional AI-powered conversational tutoring in Kinyarwanda. The ERD is designed to accommodate both traditional progress tracking and flexible AI interaction logging, allowing the platform to analyze learning patterns while maintaining a clear separation of responsibilities among system entities.

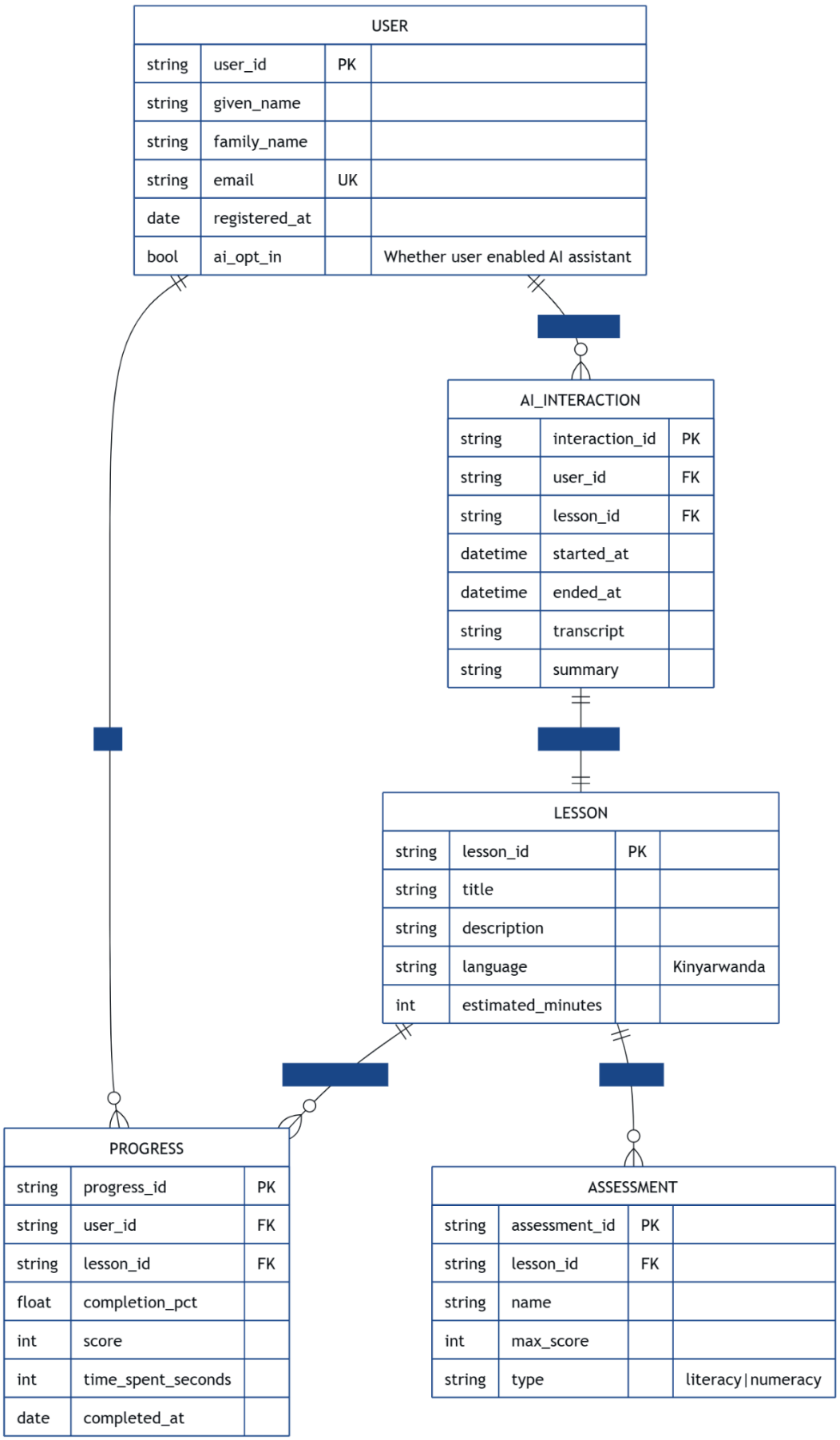
At the center of the data model is the **User** entity, which represents each adult learner participating in the platform. A single user can have multiple progress records, with each record corresponding to a specific Kinyarwanda lesson. This one-to-many relationship between User and Progress enables the system to track lesson completion status, assessment scores, time spent on lessons, and overall literacy development over time. It ensures that learner advancement is stored in a structured and retrievable manner.

In addition to tracking lesson progression, the system records optional conversational interactions through the AIInteraction entity. A single user may engage in multiple AI conversations in Kinyarwanda, depending on how often they activate the AI assistant. This establishes another one-to-many relationship between User and AIInteraction. These interactions are logged for analysis, refinement of tutoring responses, and evaluation of AI usage patterns during the pilot phase. Importantly, AI interactions are recorded only for learners who choose to activate the AI assistant, preserving the optional and user-controlled nature of conversational support.

The Lesson entity represents structured instructional units delivered in Kinyarwanda. Each lesson can be attempted by many users, forming a one-to-many relationship between Lesson and Progress. This allows the system to maintain a single standardized lesson structure while storing individualized progress data for each learner. Furthermore, every lesson is associated with one or more Assessment records designed to evaluate literacy and numeracy skills related to that specific lesson. This relationship ensures that learning outcomes are measurable and directly tied to instructional content.

Finally, AI interactions are contextualized to specific lessons through a relationship between AIInteraction and Lesson. Each conversational exchange in Kinyarwanda is linked to the lesson during which it occurred. This contextual linkage enables the system to analyze which lessons generate the most questions, identify areas where learners struggle, and continuously refine AI explanations to better support comprehension.

Together, these entity relationships create a coherent and scalable data structure that supports structured learning, individualized progress tracking, and optional AI-powered tutoring within a unified Kinyarwanda-centered system architecture.


USER
string user_id PK
string given_name
string family_name
string email UK
date registered_at
bool ai_opt_in Whether user enabled AI assistant
AI_INTERACTION
string interaction_id PK
string user_id FK
string lesson_id FK
datetime started_at
datetime ended_at
string transcript
string summary
LESSON
string lesson_id PK
string title
string description
string language Kinyarwanda
int estimated_minutes
PROGRESS
string progress_id PK
string user_id FK
string lesson_id FK
float completion_pct
int score
int time_spent_seconds
date completed_at
ASSESSMENT
string assessment_id PK
string lesson_id FK
string name
int max_score
string type literacy|numeracy

3.6 Development Tools

The system is developed using a mobile-first, cloud-connected, and largely open-source technology stack to ensure cost efficiency, scalability, and long-term sustainability within the Rwandan context. The selection of development tools prioritizes affordability, ease of maintenance, and compatibility with widely available Android devices commonly used in Kigali. This approach ensures that the platform remains technically robust while remaining realistic for deployment in low-resource environments.

The mobile application is built using React Native, a cross-platform framework that enables efficient development of Android applications through a shared codebase. This choice reduces development time and maintenance costs while ensuring compatibility with affordable Android smartphones such as Tecno, Infinix, Itel, and entry-level Samsung devices commonly used by adults in Kacyiru and Gikondo. React Native also supports multimedia integration, which is essential for delivering audio-first Kinyarwanda lessons and visual navigation elements tailored to low-literacy users.

The backend services and API layer are implemented using Node.js. This environment provides a lightweight and scalable framework for handling user authentication, lesson delivery, content management, and communication between the mobile frontend and cloud-based AI services. Node.js supports RESTful API design, enabling modular system architecture and facilitating future expansion beyond the initial pilot phase.

Conversational tutoring functionality is powered by cloud-based AI services such as the OpenAI API or the Anthropic Claude API. These services provide natural language processing capabilities that enable contextual dialogue in Kinyarwanda when learners activate the optional AI assistant. By leveraging established cloud AI providers, the system avoids the complexity and cost of developing and hosting large-scale language models locally. This ensures access to advanced conversational capabilities while maintaining operational efficiency.

Firebase is used to manage authentication, cloud data storage, and real-time synchronization. It securely stores user profiles, lesson progress records, and AI interaction logs for learners who choose to use conversational support. Firebase's managed infrastructure reduces administrative overhead while ensuring reliability and scalability.

User interface design and prototyping are conducted using Figma, allowing iterative testing of low-literacy, Kinyarwanda-only interface layouts before implementation. This supports participatory design practices and ensures the final interface aligns with the needs of adult learners. Version control and collaborative development are managed through GitHub, providing structured code management, issue tracking, and team coordination throughout the Agile development cycle.

To enhance accessibility, the platform integrates Google Cloud Text-to-Speech for high-quality Kinyarwanda voice synthesis. This technology enables clear audio instructions,

lesson narration, and reinforcement of reading exercises, which are essential for learners with minimal literacy skills.

Collectively, these technologies balance ease of development with long-term scalability and affordability. The optional cloud-based AI architecture enables sophisticated conversational tutoring in Kinyarwanda without requiring expensive local infrastructure. At the same time, because AI features are activated only when learners choose to engage them, mobile data usage remains manageable for urban Kigali users. This carefully selected technology stack, therefore, supports both innovation and practical deployment within Rwanda's economic and technological realities.

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